Seeing Reuse: Collection Routing Optimisation

Emissions-aware planning for on-demand kerbside reuse collection in Newcastle, with a mixed diesel and battery-electric fleet

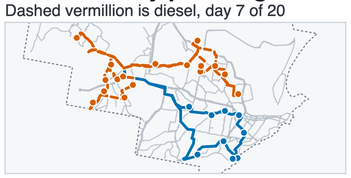
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The problem

Residents book kerbside reuse collections on demand. Each business day the planner chooses which requests to collect and routes four trucks. Distance-only planning treats every kilometre alike, yet an electric kilometre emits nothing at the tailpipe, and a loaded uphill kilometre burns more diesel than an empty downhill one.

- **RQ1** How can on-demand reuse collection be framed as a tractable green vehicle routing problem with load-dependent emissions and service deadlines?
- **RQ2** Which solver family gives the best emissions and cost trade-off at Newcastle scale?
- **RQ3** How much carbon dioxide equivalent (CO2-e) does emissions-aware planning save against distance-only planning, across demand levels and fleet mixes?

Distance-only planning
Dashed vermillion is diesel, day 7 of 20



Emissions-aware planning
Solid blue is electric. One illustrative run



The planner RQ1

1. **Requests.** 25 a day, up to 2 m3 each, synthetic, each told its collection day
2. **Road network.** OpenStreetMap roads of Newcastle, times and distances from the Open Source Routing Machine (OSRM), elevation from a 1 m laser survey averaged to 5 m
3. **Stage 1, which day and which truck.** VROOM (Vehicle Routing Open-source Optimization Machine) re-plans the next 7 business days every day. Arc cost is gradient-aware fuel from the comprehensive modal emissions model (CMEM), electric arcs near zero
4. **Stage 2, each day's routes.** Adaptive large neighbourhood search re-routes each day for fuel that depends on the load on board. Used for the three-trip figures only
5. **Check, then execute.** Capacity 11 m3, shift 05:00 to 15:00, 110 km electric planning distance, never before the advised day

How it was tested

- Both arms on identical requests, paired by seed
- 30 business days a run, 10 warm-up and 20 measured
- Most confirmatory results pre-registered in git before they ran
- Paired Wilcoxon tests, Holm correction, bootstrap 95% intervals
- Exploratory settings tuned on seeds 1 to 30, quoted on 31 to 90

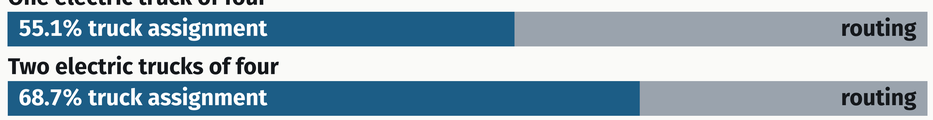
Planning for emissions instead of distance cut tailpipe CO2-e per request by more than a quarter with one electric truck of four, and by more than half with two, on the same requests and the same fleet.

−28.4%
One electric truck of four
0.725 kg CO2-e less per request

−55.8%
Two electric trucks of four
0.863 kg CO2-e less per request

Tier-1 CO2-e per request, single trip, 25 a day, 100 seeds, medians −28.35 and −55.78%, both pre-registered hypotheses hold. Electric trucks at 0 g per km, synthetic demand, 18.16 and 15.99% more km driven. **Charged on the NSW (New South Wales) grid** at 600 to 780 g per km the saving falls to 1.48 to 10.75% (0.052 to 0.313 kg), 30 seeds.
Tank to wheel. Well to wheel (upstream diesel, National Greenhouse Accounts Factors 2026) every kilogram is 24.6% higher, percentages unchanged, grid cases 1.92 to 12.78%.

Where the saving comes from



Share of the saving from giving the longest routes to the electric trucks, 55.1 to 73.7% over the six mixed cells. Routing adds the rest, smaller but significant. Post hoc, ratio of means, single trip, 30 seeds.

With up to three trips a truck a day

−82.29%
1 electric, 25 a day, 1.411 kg

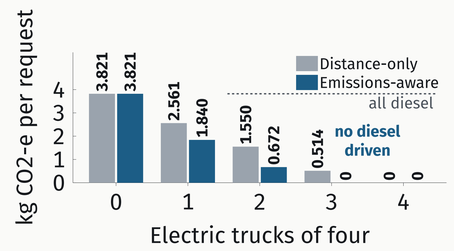
−44.42%
1 electric, 50 a day, 0.687 kg

−77.24%
2 electric, 25 a day, 0.920 kg

−83.39%
2 electric, 50 a day, 0.903 kg

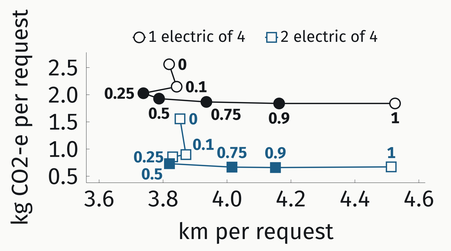
CMEM CO2-e per request with stage 2, within-model, 100 seeds, under the planner's own trip rules (unsourced). Tier-1, the independent check at 100 seeds, −79.86, −42.73, −74.62 and −81.30%.

Fleet mix exploratory



Tier-1 medians, seeds 31 to 90. Gaps are planning savings, steps are fleet decisions.

Trade-off exploratory



Approximate trade-off curve by weighted sum, among the weights tried. Labels give w, the weight on emissions. Filled points non-dominated.

The evaluation questions, answered

- **EQ1** Does it cut CO2-e? **Yes**, by 28.35 and 55.78%, centre column.
- **EQ2** At what cost? 18.16 and 15.99% more km, a mean wait 0.79 and 0.96 business days longer, 2 and 3 fewer truck-days of 80.
- **EQ3** Within 7 working days? **Yes**, every collected request in all 600 runs.
- **EQ4** Sensitive to inputs? **Direction held** in every test below.
- **EQ5** Viable against naive planning? **Partly**, 36.66% below nearest neighbour, re-plans in a median 0.57 and 0.87 s on a dedicated 4 vCPU machine, compute energy unmeasured.

EQ2 and EQ3 at 100 seeds, EQ5 at 30 seeds, its timing over 180 runs.

Does it hold? EQ4

NSW grid charging, 600 to 780 g per km	Falls to 1.48 to 10.75%
Four diesel, 15 and 25 a day, no saving	+0.66 to +0.92%
Refuse-truck diesel, 2,089 to 3,423 g per km	**Holds**, within 2 points
Low and high pick-up volumes	**Holds**, 0.619 to 0.837 kg
Clustered locations, random counts	**Holds**
Compaction, 1.0 to 2.0 times 11 m3	**Holds**
CMEM engine and fuel parameters	**Holds** in 77 of 77 cells
Driver, service time and speed cases	**Holds**, earlier objectives
Bigger bodies and demand (exploratory)	**Holds**, 17 to 100%

Single trip, 25 a day, 30 seeds unless stated, tier-1. CMEM row within-model, every campaign. Bigger bodies of 23 and 30 m3 as volume only, 25 to 100 a day, seeds 31 to 90, its overloaded 75 a day cells seeds 31 to 50.

Where it pays off exploratory

kg CO2-e less per request	1 electric	2 electric
15 a day, single trip	0.839	0.532
25 a day, up to 3 trips	2.178	1.384
50 a day, single trip, overloaded	0.493	0.507

Tier-1, fresh seeds 31 to 90. Each row changes demand or trip rules.

Was VROOM the right solver? RQ2

Every solver run in the pipeline kept the saving's direction. None clearly beat VROOM.

Solver	Plan cost against VROOM
OR-Tools guided local search	+4.83 to +7.48%
OR-Tools tabu search	+5.74 to +20.99%
OR-Tools simulated annealing	+12.26 to +28.51%
PyVRP, two releases	−0.49 (a tie) to +9.40%
Ant colony (Bullnheimer et al. 1999)	+13.94 to +108.15%
LKH-3, static days only	+0.01% (a tie)
Three learned solvers, static days	+5.58 to +32.19% distance

One untuned setting each, so this does not show VROOM is best.

Problem class. A rolling-horizon, multi-period, heterogeneous fixed fleet vehicle routing problem whose daily routes minimise load-dependent CMEM fuel at fixed network speeds, solved by a two-stage decomposition. The daily routing stage is the Pollution-Routing Problem of Bektaş and Laporte (2011) with speeds fixed and no time windows.

Reading the figures. Tier-1 CO2-e is km times a per-vehicle factor, diesel 1000 and electric 0 g per km. CMEM (Barth et al. 2005) figures are within-model. Percent and kg are separate medians of paired differences and do not convert. Tank to wheel unless stated. Exploratory, with stage 2 on both plans the same 100 seeds give −29.3 and −56.4% tier-1 CO2-e per request (stage 1, as pre-registered, −28.4 and −55.8).

Limits. Collection routing only, not reuse, item assessment or council rules. Synthetic demand and locations. Diesel factor and 110 km electric distance unsourced. 11 m3 body, no compaction. Battery manufacture out of scope, no claim of minimal emissions. Fleet sizing not reached, and the trade-off curve is approximate. Stage 2 is used for the three-trip figures only, every other figure is stage 1.